

Stony Brook University
Alejandro Melo Ponce

Fall 2016

ASSIGNMENT 4

Due: Tuesday, November 1 in class.

INSTRUCTIONS: Show all your work to get full points.

1. Suppose that the utility function for goods x and y is given by:

$$U(x, y) = (x + 1)y.$$

- Given a fixed level of utility, $\bar{U}$, graph the indifference curve. Based on your graph, explain why you might get a corner solution.
 - Calculate the uncompensated (Marshallian) demand functions for x and y . Be careful of the possibility for having a corner solution for x , i.e. $x^* = 0$ and explain what conditions need to hold so that you get a positive solution for x^* .
 - Describe how the Marshallian demand curve for x and y are shifted by changes in I or in the price of the other good.
 - Calculate the expenditure function for x and y .
 - Use the expenditure function in the previous part to compute the compensated demand functions for goods x and y . Describe how the compensated demand curves for x and y are shifted by changes in income or by changes in the price of the other good.
2. Suppose that Laura regards ham and cheese as pure complements—she will always use one slice of ham in combination with one slice of cheese to make a ham and cheese sandwich. Suppose also the ham and cheese are the only goods Laura buys and that bread is free.
- If the price of ham is equal to the price of cheese, show that the own-price elasticity of demand for ham is -0.5 and that the cross price elasticity of demand for ham with respect to the price of cheese is also -0.5 .
 - Explain why the results from the previous part reflect only income effects, not substitution effects. What are the compensated price elasticities in this problem?
 - Use the results from the previous part to show how your answers in part (a) would change if a slice of ham cost twice the price of a slice of cheese.
 - Explain how this problem could be solved intuitively by assuming Laura consumes only one good—a ham and cheese sandwich.
3. Consider a simple quasi-linear utility function of the form $U(x, y) = x + \ln(y)$.¹
- Calculate the income effect for each good. Also calculate the income elasticity of demand for each good.
 - Calculate the substitution effect for each good. Also calculate the compensated own-price elasticity of demand for each good.

¹Check the solutions for homework 2.

- (c) Show that the Slutsky equation apply to this function.
- (d) Show that the elasticity form of the Slutsky equation also applies to this function. Describe any special features you observe.